

Generalising a spatial transcriptomics analysis into a reusable R pipeline

A reproducible R framework for SpatialExperiment objects that separates data preparation from spatial clustering, so tools can be integrated and compared on identical input.



Zoe Marsland
Dr Marie Trussart
Papenfuss Lab

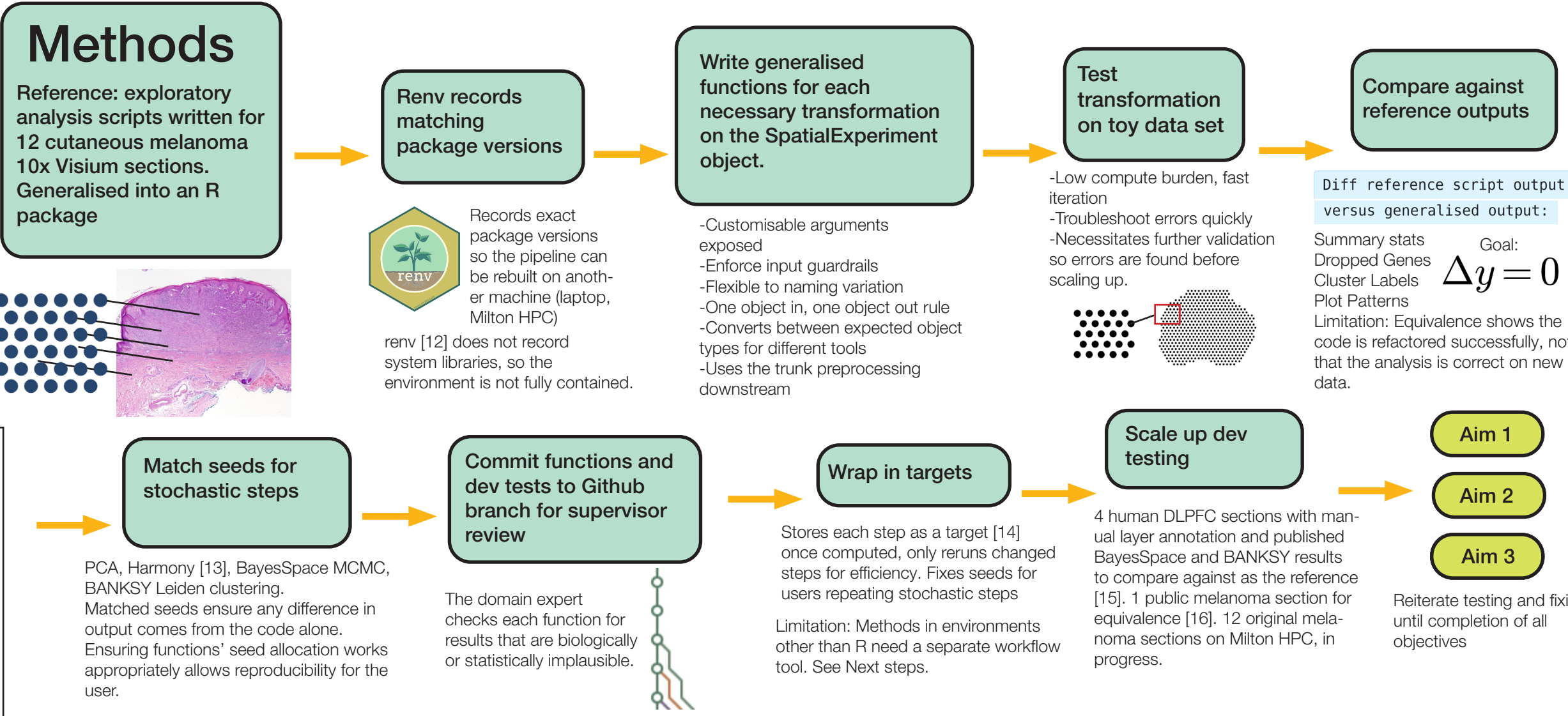
Aims

This project aims to generalise a spatial transcriptomics analysis written for one dataset into a reusable R pipeline that runs on new data without changing its results.

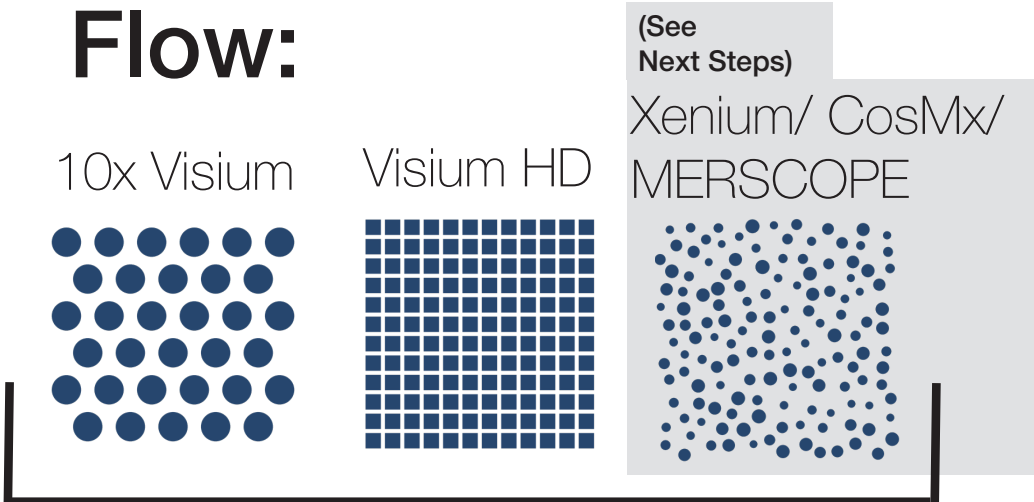
- Convert the analysis scripts into functions with adjustable thresholds, run through a targets pipeline that converts objects between clustering tools.
- Show the pipeline reproduces the original scripts' outputs on independent data.
- Show the pipeline reproduces published results for each clustering tool.

Background

Spatial transcriptomics measures the expression of thousands of genes at bar-coded spots across a tissue section, keeping the position of each measurement [1, 2]. Methods that group these spots into tissue domains are built on different data structures: SpatialExperiment [3] and SingleCellExperiment [4] in Bioconductor, the Seurat object [5], and AnnData in Scanpy [6]. Each method also applies its own normalisation and gene selection before clustering, such as scran [7] in BayesSpace [8] and Seurat's methods in BANKSY [9]. Frameworks such as Giotto [10] and Squidpy [11] combine methods within one environment, but each is built around its own structure and defaults, and none is designed to reproduce an existing analysis. Exploratory analysis scripts are also hard to reproduce. The original scripts here set no seeds for their random steps and could not reproduce their own clusters. We have built an R pipeline that runs each tool from one preprocessed object and returns every result to it, without changing the original analysis's results.



Pipeline Flow:



Targets Wrapper

Preprocessing

Removes poor quality spots (off tissue, too few reads, high mitochondrial content.)

Normalisation

Scales each spot by its total reads so spots can be compared.

Common HVGs

Keeps the most variable genes, as one shared list for all sections.

Reduced Dimensions

Harmony Batch Correction

Removes differences between sections so spots group by tissue, not sample.

Bash/Nf Wrapper (See Next Steps)

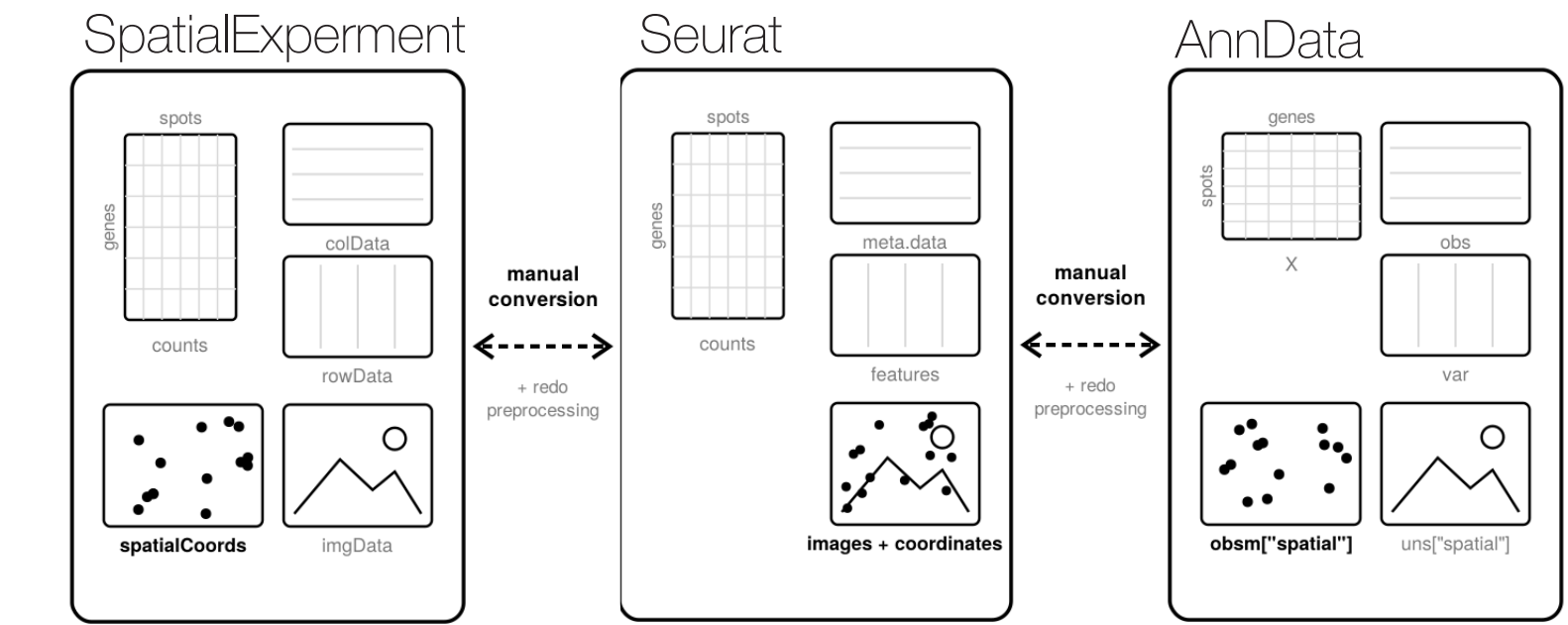


Figure 1. Data structures of three spatial clustering tools. Each tool stores counts, spot metadata and coordinates in a different place, and AnnData stores the count matrix transposed. Moving data between tools needs manual conversion and repeated preprocessing, so each tool starts from a different input.

Table 1. Manual steps against pipeline replacement

Before	With the pipeline
Each tool needs its own data structure	Object conversion is handled
Preprocessing is redone for each tool	Different clustering methods can be compared efficiently and without noise from differences in input
Settings are fixed inside scripts	Arguments can be adapted + updated to the user's data
Plots are made by hand for each sample	Plots are automated

Validation

We validated the pipeline against the original scripts and published results using the adjusted Rand index (ARI) [17]. The first comparison tests whether converting the scripts into functions altered the analysis, and the second tests whether the pipeline generalises to other tissue. On a public melanoma section [16] ($n = 2,599$ spots), all three tools matched the original scripts ($ARI = 1$), as did the top 50 marker genes (Fig. 2, 4). Changing one parameter reduced agreement to $ARI = 0.77$. On DLPFC section 151673 [15], BayesSpace and BANKSY each came within 0.01 of their published ARI (Fig. 3).

Equivalence Test

Original reference script clustering vs pipeline functions on same parameters and new melanoma data [16].

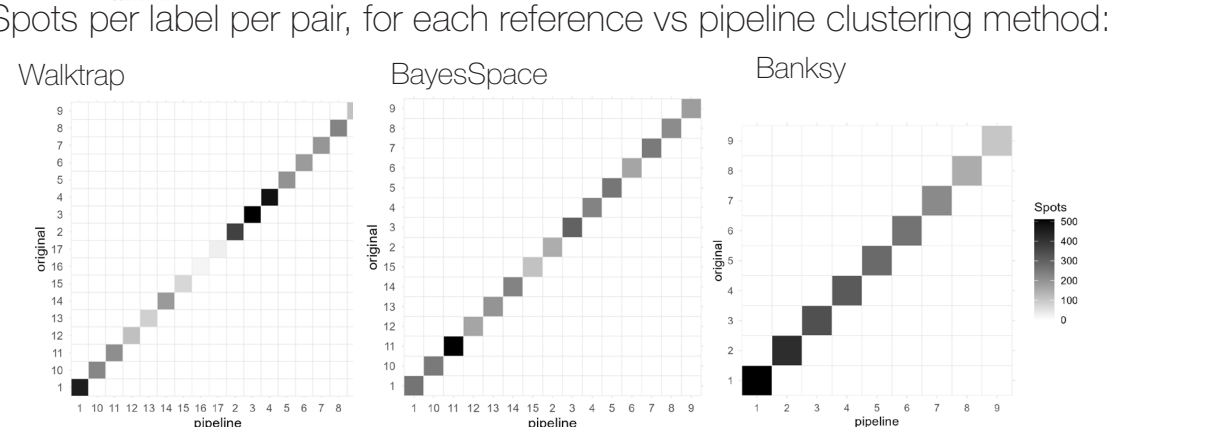
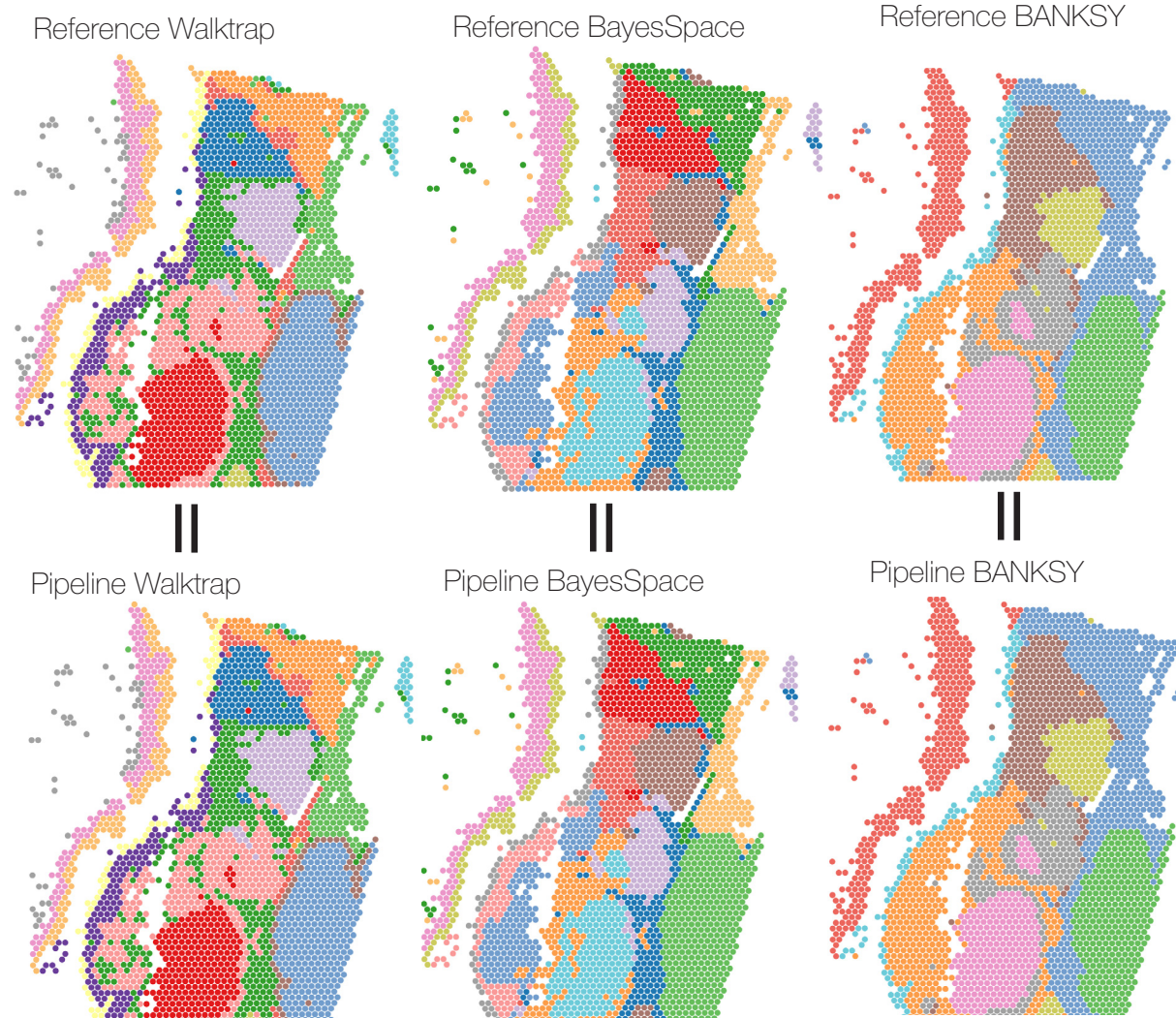


Figure 2. Original scripts and the pipeline on a public melanoma section [16] (a) Cluster maps from each tool are identical between the original scripts (top) and the pipeline (bottom). (b) Number of spots for each pair of labels. Every spot lies on the diagonal, so each original cluster maps to a single pipeline cluster.

Next steps

- Further testing**
Test equivalence on the original cohort of 12 samples on Milton HPC (in progress).
Test batch correction and joint clustering on several DLPFC sections, with and without the coordinate offset.
Rerun stochastic steps across several seeds to give a reference range for differences.
Test the pipeline on Xenium [18], an in situ platform that measures single cells against a targeted gene panel. It differs from Visium in resolution, gene coverage and data size.
- Analyses after clustering**
Add cell type deconvolution with RCTD [19] as a pipeline step. It is currently run separately on the clustered object.
Test differences in cluster composition between patient groups with sccomp [20].
Add spatial interaction analysis between cell types with SPIAT [21].
- Environments**
Convert to Seurat and AnnData, so tools in Python can be used without manual export.
Port the workflow to Nextflow [22] for scheduling across samples on HPC, targets manages steps within R but not jobs across a cluster.

Clustering: Reference vs Pipeline

DLPFC section 151673: published (bottom) and pipeline (top)

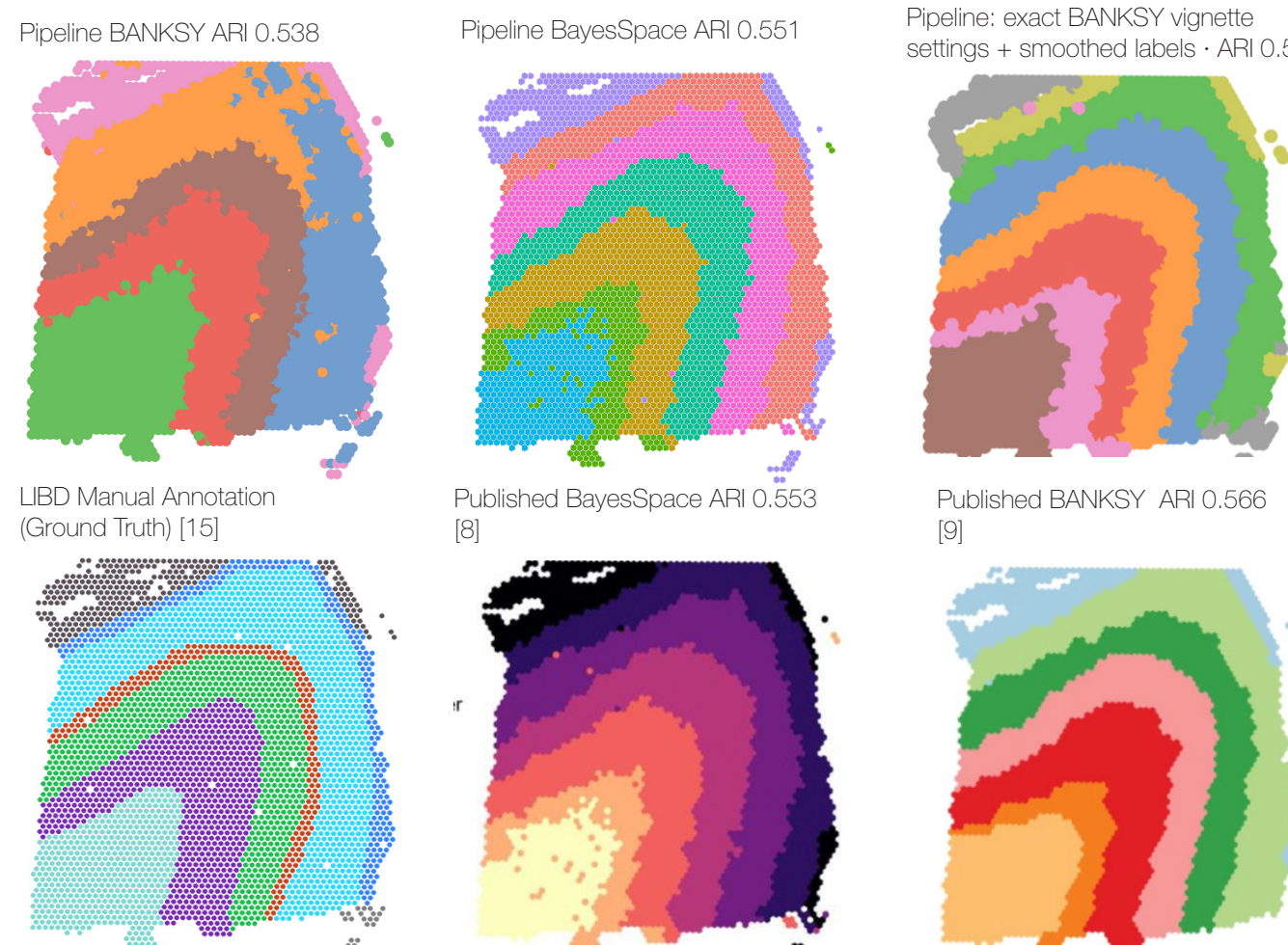


Figure 3. Clustering of DLPFC section 151673 by the pipeline (top) and published results (bottom). Manual layer annotation [15] is the reference for ARI. All three pipeline panels recover the layered bands of the cortex. BayesSpace matched its published ARI using arguments alone. BANKSY came within 0.01 only after functions for the published gene selection and label smoothing were added, so its published result depends on preprocessing the original analysis did not use.

Top Markers: Reference vs Pipeline

Marker genes ranked per BANKSY cluster by median Cohen's d in the original script and the pipeline returned the identical heatmap:

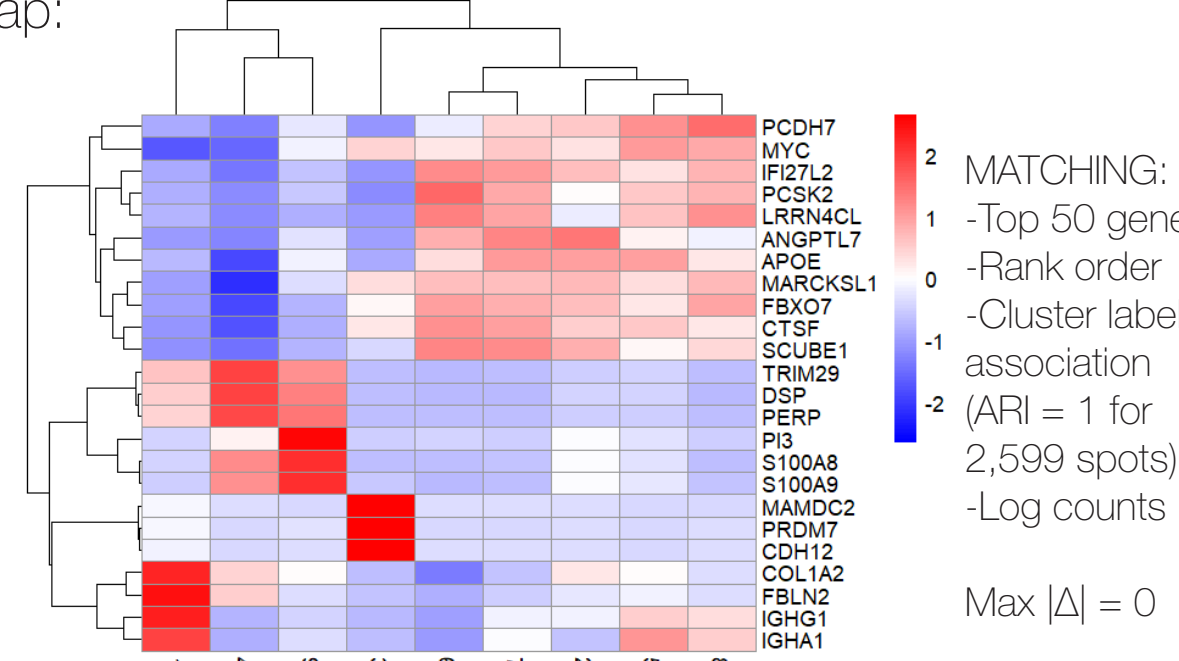


Figure 4. Top marker genes per BANKSY cluster on the melanoma section, ranked by median Cohen's d . Colour shows scaled log expression. The original script and the pipeline returned the same genes in the same order with no difference in values ($\max |\Delta| = 0$), so results after clustering are also reproduced.

Conclusion

- 38**
functions replace the original analysis scripts
- ARI = 1**
for Walktrap, BayesSpace and BANKSY against the original scripts on new data
- 0.551**
BayesSpace on brain tissue, against 0.553 published

We converted a Visium analysis into 38 functions run through a targets pipeline without altering its output. On a public melanoma section ($n = 2,599$ spots), the pipeline and the original scripts assigned all spots to the same clusters for Walktrap, BayesSpace and BANKSY ($ARI = 1$) and returned the same marker genes, while changing k from 10 to 11 reduced agreement to $ARI = 0.77$ (Fig. 2, 4). On DLPFC section 151673, the pipeline reproduced the published BayesSpace result through arguments alone ($ARI = 0.551$ and 0.553, respectively). BANKSY approached its published result only after its own gene selection and smoothing were added ($ARI = 0.558$ and 0.566) (Fig. 3). These comparisons used one seed each, so differences of this size cannot yet be separated from seed variation. The results suggest that differences reported between spatial clustering methods may reflect their preprocessing as well as the methods, motivating comparison from a shared object. Batch correction and joint clustering across samples remain to be tested.

Acknowledgements

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Citations

